Generative Models in Finance

Week 3: LLMs in Finance and Investment Management

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Spring 2026

Overview

1. The Information Gap
2. Sentiment Analysis
3. Financial Time Series
4. LLM Agents in Finance
5. Challenges and Takeaways

Reference: Y. Kong, Y. Nie, X. Dong, J. M. Mulvey, H. V. Poor, Q. Wen & S. Zohren, *Large Language Models for Financial and Investment Management: Applications and Benchmarks*, Journal of Portfolio Management, 2024.

Reading Between the Lines

> *"Recent indicators suggest that economic activity has continued to expand at a solid pace. Job gains have slowed, and the unemployment rate has moved up but remains low. Inflation has made further progress toward the Committee's 2 percent objective but remains somewhat elevated."*
>
> — FOMC Statement, September 2024

- This paragraph moved Treasury yields within minutes of release
- The language is deliberately calibrated: "solid pace" signals confidence, "somewhat elevated" signals caution. The precise choice of words *is* the message
- A macro strategist reads these nuances instantly — but cannot read 10,000 documents per day
- Question: can a machine extract the same signal from text, at scale?

The Scale of the Problem

- Quantitative finance has traditionally relied on structured data: prices, volumes, balance sheet items — numbers in rows and columns
- But the vast majority of financial information is unstructured text:
  - 10-K filings: $\sim$50,000 words each $\times$ $\sim$4,000 US public companies
  - Earnings calls: $\sim$3,000 per quarter, each 30–60 minutes
  - News: thousands of articles per day from Reuters, Bloomberg, the FT
  - Social media: millions of daily posts on Twitter/X and StockTwits
  - Central bank communications: FOMC minutes, ECB statements, speeches
- A single equity analyst covers $\sim$15 stocks. The gap between available text and human capacity is enormous
- LLMs are the first technology that can process unstructured text at this scale and extract structured, quantitative signals. The rest of this lecture examines how — and where they fall short

What Is Sentiment Analysis?

- Sentiment analysis assigns a score (positive, negative, or neutral) to a piece of text, quantifying its opinion or tone
- Financial application: given a news headline, an earnings transcript, or a central bank statement, estimate how *positive* or *negative* the market-relevant content is
- Why it matters: if sentiment can be measured *before* the market fully reacts, it becomes a trading signal
- Core sub-tasks in the literature:
  - Scoring news headlines and social media posts
  - Analysing tone in corporate disclosures and earnings calls
  - Interpreting central bank communications (FOMC minutes, ECB statements)

From Word Counts to Contextual Understanding

- Layer 1 — Dictionaries. Count positive/negative words from predefined lists. The Loughran–McDonald (2011) dictionary is finance-specific: "liability" is negative in everyday English but neutral in a filing. Simple and interpretable, but blind to negation and context (e.g. "no significant decline in margins" is flagged negative because of "decline")
- Layer 2 — Static embeddings. Word2Vec, GloVe: represent words as vectors in $\mathbb{R}^d$. Capture semantic similarity, but produce the same vector for "bank" whether it means a financial institution or a river bank
- Layer 3 — Contextualised models (BERT onward): every token's representation depends on its full surrounding context. The model distinguishes word senses and recognises negation. This is the key transition: from counting words to understanding sentences

Loughran, T. & McDonald, B. (2011). When Is a Liability Not a Liability? *Journal of Finance* 66(1): 35–65.

Finance-Specific Language Models

- FinBERT (Araci 2019): BERT (110M parameters) fine-tuned on financial news. Learns domain-specific word senses. The most widely used baseline for financial sentiment
- BloombergGPT (Wu et al. 2023): 50 billion parameters, pre-trained from scratch on 363B tokens of Bloomberg's proprietary financial text. Outperforms general models on financial NLP while remaining competitive on general benchmarks
- General-purpose models (GPT-4, Claude, LLaMA): not finance-specific, but often competitive with or superior to fine-tuned models, especially on tasks requiring broad world knowledge
- A note on terminology: the literature uses "LLM" broadly, from BERT (110M params) to GPT-4 (>1T). The capability gap is substantial — keep this in mind when interpreting results

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Wu, S. et al. (2023). BloombergGPT: A Large Language Model for Finance. arXiv:2303.17564.
Araci, D. (2019). FinBERT: Financial Sentiment Analysis with Pre-Trained Language Models. arXiv:1908.10063.

Does LLM Sentiment Predict Returns?

- Lopez-Lira & Tang (2023): GPT-4 scores a single news headline → predicts the *next-day* cross-sectional stock return. A daily long/short strategy achieves a Sharpe ratio of ∼3.8. Effect strongest for small caps and after negative news — consistent with slow information diffusion
- Chen, Kelly & Xiu (2022): use BERT/RoBERTa embeddings of news articles to form daily-rebalanced portfolios → annualised Sharpe ratio of ∼4.8 (2004–2019), significantly above Word2Vec baselines. Prices respond to news with a multi-day delay, creating a short-term trading window
- Caveat: both studies report in-sample or simulated results with high turnover. Transaction costs, slippage, and out-of-sample decay can erode these Sharpe ratios substantially

Lopez-Lira, A. & Tang, Y. (2023). Can ChatGPT Forecast Stock Price Movements? arXiv:2304.07619.
Chen, Y., Kelly, B.T. & Xiu, D. (2022). Expected Returns and Large Language Models. SSRN 4416687.

How Fragile Is the Signal?

- Adversarial fragility: Leippold (2023) replaces a few words with synonyms ("revenue grew" → "revenue expanded") and re-queries GPT-3. These minimal, meaning-preserving edits flip the sentiment score
- The Loughran–McDonald dictionary is *immune* to this: synonyms map to the same polarity. LLM sophistication introduces a new attack surface that simpler methods do not have
- Strategic adaptation: Cao et al. (2023) document that corporations are already adjusting the language of their disclosures, knowing that AI systems read them. This is a new form of strategic communication aimed at *machine* readers — a feedback loop between AI and corporate disclosure
- Implication: if a competitor or bad actor can manipulate the text inputs to your LLM-based strategy (crafted press releases, social media posts), the resulting trading signals can be systematically corrupted

Leippold, M. (2023). Sentiment Spin: Attacking Financial Sentiment with GPT-3. *Finance Research Letters* 55: 103957.
Cao, S. et al. (2023). How to Talk When a Machine Is Listening. *Rev. Financial Studies* 36(9): 3603–3642.

Sentiment: Limits and Verdict

- Multimodal gaps: Curti & Kazinnik (2023) find that *nonverbal* cues in FOMC press conferences (facial expressions, tone of voice) carry significant information about monetary policy. Text-only LLMs miss this entirely
- Social media noise: Twitter/X and StockTwits contain misinformation, sarcasm, and coordinated manipulation campaigns. LLMs may not distinguish genuine sentiment from deliberate deception
- Out-of-sample fragility: most studies test on 1–3 years of US equity data. Persistence across market regimes, asset classes, and geographies remains largely untested
- Signal decay: if many participants deploy similar LLM sentiment strategies, the arbitrage opportunity should shrink as markets incorporate information faster
- Verdict: LLM sentiment is a promising *input* to a broader quantitative model, not a standalone trading strategy. Robustness testing — across time, assets, and adversarial conditions — is essential before deployment

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Curti, F. & Kazinnik, S. (2023). Let's Face It: Quantifying the Impact of Nonverbal Communication in FOMC Press Conferences. *J. Monetary Economics* 139: 110–126.

Can Language Models Understand Numbers?

- LLMs are trained on *text*. A financial time series is a sequence of *numbers*. At first glance, these have nothing in common
- But both are sequential data: a sentence is a sequence of tokens, a price series is a sequence of values. The transformer architecture processes sequences — it does not inherently care whether the tokens represent words or digits
- Key insight: the web text that LLMs are pre-trained on contains vast amounts of numerical sequences — tables of stock prices, macroeconomic data in Wikipedia, financial statements in HTML. The model has already seen numbers in temporal context

Zero-Shot Forecasting with LLMs

- Key idea (Gruver et al. 2023): take a pre-trained LLM (GPT-3, LLaMA-2) with *no* time-series training. Convert a numerical series into a text string — e.g. `"1423.5, 1418.2, 1425.7, ..."` — and prompt the model to continue the sequence
- Why does this work at all? LLMs are trained on vast text corpora that include tables, financial reports, and numerical data. They learn implicit patterns: trends, seasonality, and repetition. Their ability to represent *multimodal distributions* over next tokens lets them capture uncertainty naturally
- Result: on standard benchmarks (Darts, Monash), zero-shot GPT-3 matches or exceeds purpose-built models trained on each dataset. This is surprising — the model was never explicitly trained to forecast
- Limitations: (i) tokenisation splits numbers into character fragments, so the model has no native sense of numerical magnitude; (ii) RLHF-aligned models (e.g. GPT-4) perform *worse* than base models, likely because alignment distorts the output distribution; (iii) performance degrades on noisy, low signal-to-noise series typical of finance

Gruver, N. et al. (2023). Large Language Models Are Zero-Shot Time Series Forecasters. *NeurIPS* 36: 19622–19635.

Time-Series Foundation Models

- Gruver et al. showed that LLMs can forecast zero-shot, but repurposing a *language* model for *numerical* data is inefficient. This has motivated purpose-built foundation models for time series: pre-train a transformer once on large-scale time-series data, then apply it zero-shot to new forecasting problems
- TimesFM (Das et al. 2024, Google): decoder-only transformer pre-trained on $>$100B real-world time points. Produces *point forecasts*. Zero-shot performance competitive with supervised per-dataset models on standard benchmarks
- Chronos (Ansari et al. 2024, Amazon): repurposes the T5 (encoder–decoder) architecture. Tokenises continuous values via scaling and uniform quantisation into discrete bins, turning forecasting into a language-modelling problem over bin indices
- Moirai (Woo et al. 2024, Salesforce): encoder-only transformer that outputs *probabilistic* forecasts via mixture distributions — providing prediction intervals, not just point estimates. Handles multivariate input natively
- Open question for finance: these models are evaluated primarily on smooth, high signal-to-noise benchmarks (weather, electricity, traffic). Financial time series are noisier, non-stationary, and adversarial — performance on financial data remains largely untested

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Das, A. et al. (2024). A Decoder-Only Foundation Model for Time-Series Forecasting. *ICML* 2024.
Ansari, A.F. et al. (2024). Chronos: Learning the Language of Time Series. arXiv:2403.07815.
Woo, G. et al. (2024). Unified Training of Universal Time Series Forecasting Transformers. *ICML* 2024.

Reality Check: LLMs vs. Traditional Models

- Xie et al. (2023): systematic zero-shot evaluation — ChatGPT underperforms both traditional ML (gradient-boosted trees, SVMs) and state-of-the-art deep learning on stock movement prediction
- Why? Two fundamental problems:
  1. Numerical precision: tokenising "1423.50" as a string of characters loses magnitude — the model has no native understanding that 1423 is close to 1400 and far from 14
  2. Distributional shift: financial series are non-stationary. An LLM pre-trained on historical text has no mechanism to adapt to regimes it has never seen
- The efficient market objection: if a publicly available LLM can predict returns from publicly available news, the signal should be arbitraged away rapidly. Persistent predictability would imply a market inefficiency hard to reconcile with decades of asset pricing research
- Even the foundation models (TimesFM, Chronos, Moirai) are primarily evaluated on *general* benchmarks (weather, electricity, traffic). Financial series — noisier, more adversarial, lower signal-to-noise — are less tested

Xie, Q. et al. (2023). The Wall Street Neophyte: Zero-Shot Analysis of ChatGPT over Multimodal Stock Movement Prediction. arXiv:2304.05351.

Time Series: Pitfalls and Verdict

- Look-ahead contamination: many studies use LLMs trained *after* the test period. The model may have memorised events or patterns from its training data that overlap with the backtest window — a critical methodological flaw that is easy to miss and hard to rule out
- Reproducibility: commercial APIs (GPT-3.5, GPT-4) change silently over time. A study run in January 2023 may not reproduce in June 2023. Rigorous out-of-sample testing is difficult when the model is a black box that updates without notice
- Verdict: LLMs are most convincing as feature extractors — generating text-derived signals (sentiment, entity relationships, summaries) that feed into classical quantitative models (regression, portfolio optimisation). Using an LLM as a direct price forecaster remains speculative and under-evidenced

Yu, X. et al. (2023). Temporal Data Meets LLM. arXiv:2306.11025.
Chen, Z. et al. (2023). ChatGPT Informed GNN for Stock Movement Prediction. arXiv:2306.03763.

What Is an LLM Agent?

- A standard LLM is reactive: you give it a prompt, it returns a response, the interaction ends
- An LLM agent adds three capabilities that turn it into an autonomous system:
  1. Tool use: the model can call external functions — fetch a stock price, query a database, execute a trade via an API, run Python code
  2. Planning: the model breaks a complex goal into sub-steps and executes them sequentially, deciding at each step what to do next
  3. Memory: the model maintains persistent state across interactions — it remembers past actions and outcomes
- The agent operates in a loop: perceive the environment $\rightarrow$ reason $\rightarrow$ act $\rightarrow$ observe the result $\rightarrow$ repeat
- In finance: read today's market data and news $\rightarrow$ form a trading thesis $\rightarrow$ place an order $\rightarrow$ observe the P&L $\rightarrow$ update the strategy

From Simple Bots to Autonomous Traders

- The simplest trading agent: prompt an LLM with today's news, ask "buy or sell?".
  Problem: no memory — each query starts from scratch, no learning from past trades
- FINMEM (Yu et al. 2024): working memory + layered long-term memory (past trades, outcomes, lessons) prioritised by relevance and recency. The agent evolves its strategy through continuous learning
- FinAgent (Zhang, Zhao et al. 2024): the first *multimodal* trading agent — text, numerical data, and candlestick chart images via GPT-4V. Outperforms FINMEM by adding visual pattern recognition
- QuantAgent (Wang, Yuan et al. 2024): self-improving agent with two loops — an inner loop refines strategies via a knowledge base, an outer loop validates on real market data. Discovers trading signals autonomously
- Alpha-GPT 2.0 (Yuan et al. 2024): human-in-the-loop — a trader proposes alpha ideas, the agent formalises, backtests, and refines them iteratively

Yu, Y.H. et al. (2024). FinMem. *AAAI Symposium Series*, Vol. 3, pp. 595–597.
Zhang, W. et al. (2024). FinAgent. arXiv:2402.18485.
Wang, S. et al. (2024). QuantAgent. arXiv:2402.03755. Yuan, H. et al. (2024). Alpha-GPT 2.0. arXiv:2402.09746.

Multi-Agent Systems and Market Simulation

- TradingAgents (Xiao et al. 2024): mimics the structure of a real trading firm — specialised LLM agents act as fundamental analysts, sentiment analysts, technical analysts, and traders. Agents debate and synthesise reports before the trader decides
- StockAgent (Zhang, Liu et al. 2024): populates a simulated market with LLM agents. Removing information channels (bulletin boards) significantly alters price dynamics — revealing how information flow drives price formation. This connects directly to market microstructure theory
- "Homo Silicus" (Horton 2023): LLM agents endowed with behavioural-economics preferences simulate human economic decisions. These "silicon subjects" can replace costly human experiments in social science
- No strong evidence that any LLM agent consistently generates alpha in *live* markets. All results are simulated or backtested
- Verdict: multi-agent systems are powerful for *research* (understanding market dynamics) and *augmentation* (assisting human traders). Fully autonomous LLM trading remains speculative

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Xiao, Y. et al. (2024). TradingAgents: Multi-Agents LLM Financial Trading Framework. arXiv:2412.20138.
Zhang, C. et al. (2024). Stockagent. arXiv:2407.18957. Horton, J.J. (2023). LLMs as Simulated Economic Agents. NBER.

Hallucination and Bias

- Hallucination: LLMs generate text that is fluent, confident, and factually wrong — they optimise for plausibility, not truth. In finance, a hallucinated revenue figure or a fabricated analyst rating is indistinguishable from a real one. Unlike a search engine that returns "no results," an LLM *always* produces an answer
- Mitigation: Retrieval-Augmented Generation (ground answers in source documents), code-based reasoning (compute rather than generate numbers), multi-agent verification
- Bias: Chuang & Yang (2022) find BERT and FinBERT systematically prefer certain stocks (e.g. Tesla) over others (e.g. Ford) in fill-in-the-blank prompts. An investment system built on these embeddings inherits these biases as systematic tilts
- Fairness: Lakkaraju et al. (2023) test ChatGPT on financial questions across English, AAVE, and Telugu. The models give inconsistent recommendations across demographics — different advice based solely on dialect
- Niszczota & Abbas (2023): GPT-4 scores near-perfect on financial literacy tests, but users with *lower* financial knowledge rely on it *more heavily* — precisely the population most vulnerable to errors

Chuang, C. & Yang, Y. (2022). Buy Tesla, Sell Ford. *ACL*, pp. 100–105.
Lakkaraju, K. et al. (2023). LLMs for Financial Advisement. *ACM AI in Finance*, pp. 100–107.
Sarmah, B. et al. (2023). Towards Reducing Hallucination in Extracting Information from Financial Reports. *Int. Conf. on AI-ML Systems*.

Regulation and Liability

- The core question: if an LLM gives financial advice and the client loses money, who is liable — the LLM provider, the firm that deployed it, or the individual who acted on it?
- Existing frameworks were not designed for AI:
  - *Fiduciary duty* requires advisors to act in the client's best interest — but an LLM has no concept of "interest"
  - *Suitability requirements* mandate that advice match the client's risk profile — but an LLM may not reliably distinguish a retiree from a day trader
  - *Explainability* (e.g. MiFID II): firms must explain *why* a recommendation was made — chain-of-thought reasoning may or may not satisfy this standard
- Cao et al. (2023) document that corporations are adapting disclosures to be more "machine-readable" — a feedback loop between AI systems and corporate communication
- No jurisdiction has yet established clear rules for AI-generated financial advice. This regulatory vacuum is a risk for both firms and consumers

Caspi, I. et al. (2023). Generative AI and the Future of Financial Advice Regulation. GenLaw Center.
Cao, S. et al. (2023). How to Talk When a Machine Is Listening. *Rev. Financial Studies* 36(9): 3603–3642.

Three Takeaways

1. LLMs unlock unstructured financial text at scale — this is real and valuable. The vast majority of financial information was previously inaccessible to systematic strategies. LLMs make it analysable for the first time
2. They are best used as feature extractors, not standalone forecasters. The strongest evidence supports LLM-derived sentiment as an *input* to portfolio construction. Using an LLM as a direct price predictor or autonomous trader remains speculative
3. Robustness is the binding constraint on deployment. Adversarial fragility, out-of-sample instability, hallucination, bias, and regulatory uncertainty all limit real-world use. Hybrid systems — LLM reasoning combined with classical risk models and rigorous statistical testing — are the most promising path forward